

Nursing Exam Cram Sheet for the NCLEX-RN

The final mountain that nursing students must summit before becoming a registered nurse is the NCLEX. Preparing for the NCLEX can be stressful as taking in colossal amounts of information has never been easy. This is where this cram sheet can help-- it contains condensed facts about the licensure exam and key nursing information. When exam time comes, you can write and transfer these vital information from your head to a blank sheet of paper provided by the testing center.

Busy Bee Nurses, Inc.

NCLEX STUDY TIPS MADE SIMPLE

<http://www.busybee.com/learn/flashcards>

- **Rephrase the question**—putting the question into your own words can pluck the unneeded info and reveal the core of the stem.
- **Make an educated guess**—if you can't make the best answer for a question after carefully reading it, choose the answer with the most information.

2. Vital Signs

- **Heart rate:** 80–100 bpm
- **Respiratory rate:** 12-20 rpm
- **Blood pressure:** 110-120/60 mmHg
- **Temperature:** 37 °C (98.6 °F)

3. Hematology values

- **RBCs:** 4.5–5.0 million
- **WBCs:** 5,000–10,000
- **Platelets:** 200,000–400,000
- **Hemoglobin (Hgb):** 12–16 gm (female); 14–18 gm (male).
- **Hematocrit (Hct):** 37–47 (female); 40–54 (male)

4. Serum electrolytes

- **Sodium:** 135–145 mEq/L
- **Potassium:** 3.5–5.5 mEq/L
- **Calcium:** 8.5–10.9 mEq/L
- **Chloride:** 95–105 mEq/L
- **Magnesium:** 1.5–2.5 mEq/L ✓
- **Phosphorus:** 2.5–4.5 mEq/L

5. ABG Values

- **pH:** 7.36–7.45
- **HCO3:** 24–26 mEq/L
- **CO2:** 35–45 mEq/L
- **PaO2:** 80%–100%
- **SaO2:** >95%

6. Acid-Base Balance

- Remember ROME (respiratory opposite/metabolic equal) to remember that in respiratory acid/base disorders the pH is opposite to the other components.
- Use the Tic-Tac-Toe Method for interpreting ABGs. Read more about it [here \(http://bit.ly/abgtictactoe\)](http://bit.ly/abgtictactoe).

7. Chemistry Values

- **Glucose:** 70–110 mg/dL
- **Specific Gravity:** 1.010–1.030
- **BUN:** 7-22 mg/dL
- **Serum creatinine:** 0.6–1.35 mg/dL
- **LDH:** 100-190 U/L
- **Protein:** 6.2–8.1 g/dL
- **Albumin:** 3.4–5.0 g/dL
- **Bilirubin:** <1.0 mg/dL
- **Total Cholesterol:** 130–200 mg/dL
- **Triglyceride:** 40–50 mg/dL
- **Uric acid:** 3.5–7.5 mg/dL
- **CPK:** 21-232 U/L

8. Therapeutic Drug Levels

- **Carbamazepine (Tegretol):** 4–10 mcg/ml
- **Digoxin (Lanoxin):** 0.8–2.0 ng/ml
- **Gentamycin (Garamycin):** 5–10 mcg/ml (peak), <2.0 mcg/ml (valley)
- **Lithium (Eskalith):** 0.8–1.5 mEq/L
- **Phenobarbital (Solfoton):** 15–40 mcg/mL
- **Phenytoin (Dilantin):** 10–20 mcg/dL
- **Theophylline (Aminophylline):** 10–20 mcg/dL
- **Tobramycin (Tobrex):** 5–10 mcg/mL (peak), 0.5–2.0 mcg/mL (valley)
- **Valproic Acid (Depakene):** 50–100 mcg/ml
- **Vancomycin (Vancocin):** 20–40 mcg/ml (peak), 5 to 15 mcg/ml (trough)

9. Anticoagulant therapy

- **Sodium warfarin (Coumadin) PT:** 10–12 seconds (control). The antidote is Vitamin K.
- **INR (Coumadin):** 0.9–1.2
- **Heparin PTT:** 30–45 seconds (control). The antidote is protamine sulfate.
- **APTT:** 23.3–31.9 seconds
- **Fibrinogen level:** 203–377 mg/dL

10. Conversions

- 1 teaspoon (t) = 5 ml
- 1 tablespoon (T) = 3 t = 15 ml
- 1 oz = 30 ml
- 1 cup = 8 oz
- 1 quart = 2 pints
- 1 pint = 2 cups
- 1 grain (gr) = 60 mg

- 1 gram (g) = 1,000 mg
- 1 kilogram (kg) = 2.2 lbs
- 1 lb = 16 oz
- Convert C to F: C+40 multiply by 9/5 and subtract 40
- Convert F to C: F+40 multiply by 5/9 and subtract 40

11. Maternity Normal Values

- **Fetal Heart Rate:** 120–160 bpm
- **Variability:** 6–10 bpm
- **Amniotic fluid:** 500–1200 ml
- **Contractions:** 2–5 minutes apart with duration of < 90 seconds and intensity of <100 mmHg.
- **APGAR Scoring:** Appearance, Pulses, Grimace, Activity, Reflex Irritability. Done at 1 and 5 minutes with a score of 0 for absent, 1 for decreased, and 2 for strongly positive. Scores 7 and above are generally normal, 4 to 6 fairly low, and 3 and below are generally regarded as critically low.
- **AVA:** The umbilical cord has two arteries and one vein.

12. STOP—Treatment for maternal hypotension after an epidural anesthesia:

- Stop infusion of Pitocin.
- Turn the client on her left side.
- Administer oxygen.
- If hypovolemia is present, push IV fluids.

13. Pregnancy Category of Drugs

- Category A—No risk in controlled human studies
- Category B—No risk in other studies. Examples: Amoxicillin, Cefotaxime.

- Category C—Risk not ruled out. Examples: Rifampicin (Rifampin), Theophylline (Theolair).
- Category D—Positive evidence of risk. Examples: Phenytoin, Tetracycline.
- Category X—Contraindicated in Pregnancy. Examples: Isotretinoin (Accutane), Thalidomide (Immunoprin), etc.
- Pregnancy Category N—Not yet classified

14. Drug Schedules

- **Schedule I**—no currently accepted medical use and for research use only (e.g., heroin, LSD, MDMA).
- **Schedule II**—drugs with high potential for abuse and requires written prescription (e.g., Ritalin, hydromorphone (Dilaudid), meperidine (Demerol), and fentanyl).
- **Schedule III**—requires new prescription after six months or five refills (e.g., codeine, testosterone, ketamine).
- **Schedule IV**—requires new prescription after six months (e.g., Darvon, Xanax, Soma, and Valium).
- **Schedule V**—dispensed as any other prescription or without prescription (e.g., cough preparations, Lomotil, Motofen).

15. Medication Classifications

- **Antacids**—reduces hydrochloric acid in the stomach.
- **Antianemics**—increases blood cell production.

- **Anticholinergics**—decreases oral secretions.
- **Anticoagulants**—prevents clot formation,
- **Anticonvulsants**—used for management of seizures and/or bipolar disorders.
- **Antidiarrheals**—decreases gastric motility and reduce water in bowel.
- **Antihistamines**—block the release of histamine.
- **Antihypertensives**—lower blood pressure and increases blood flow.
- **Anti-infectives**—used for the treatment of infections,
- **Bronchodilators**—dilates large air passages in asthma or lung diseases (e.g., COPD).
- **Diuretics**—decreases water/sodium from the Loop of Henle.
- **Laxatives**—promotes the passage of stool.
- **Miotics**—constricts the pupils.
- **Mydriatics**—dilates the pupils.
- **Narcotics/analgesics**—relieves moderate to severe pain.

16. Rules of nines for calculating Total Body Surface Area (TBSA) for burns

- Head: 9%
- Arms: 18% (9% each)
- Back: 18%
- Legs: 36% (18% each)
- Genitalia: 1%

17. Medications

- **Digoxin (Lanoxin)**—Assess pulses for a full minute, if less than 60 bpm hold

dose. Check digitalis and potassium levels.

- **Aluminum Hydroxide (Amphojel)**—Treatment of GERD and kidney stones. WOF constipation.
- **Hydroxyzine (Vistaril)**—Treatment of anxiety and itching. WOF dry mouth.
- **Midazolam (Versed)**—given for conscious sedation. WOF respiratory depression and hypotension.
- **Amiodarone (Cordarone)**—WOF diaphoresis, dyspnea, lethargy. Take missed dose any time in the day or to skip it entirely. Do not take double dose.
- **Warfarin (Coumadin)**—WOF for signs of bleeding, diarrhea, fever, or rash. Stress importance of complying with prescribed dosage and follow-up appointments.
- **Methylphenidate (Ritalin)**—Treatment of ADHD. Assess for heart related side-effects and reported immediately. Child may need a drug holiday because the drug stunts growth.
- **Dopamine**—Treatment of hypotension, shock, and low cardiac output. Monitor ECG for arrhythmias and blood pressure.
- **Rifampicin**—causes red-orange tears and urine.
- **Ethambutol**—causes problems with vision, liver problem.
- **Isoniazid**—can cause peripheral neuritis, take vitamin B6 to counter.

18. Developmental Milestones

- **2–3 months:** able to turn head up, and can turn side to side. Makes cooing or gurgling noises and can turn head to sound.
- **4–5 months:** grasps, switch and roll over tummy to back. Can babble and can mimic sounds.
- **6–7 months:** sits at 6 and waves bye-bye. Can recognize familiar faces and knows if someone is a stranger. Passes things back and forth between hands.
- **8–9 months:** stands straight at eight, has favorite toy, plays peek-a-boo.
- **10–11 months:** belly to butt.
- **12–13 months:** twelve and up, drinks from a cup. Cries when parents leave, uses furniture to cruise.

19. Cultural Considerations

- **African Americans**—May believe that illness is caused by supernatural causes and seek advice and remedies from faith healers; they are family oriented; have higher incidence of high blood pressure and obesity; high incidence of lactose intolerance with difficulty digesting milk and milk products.
- **Arab Americans**—May remain silent about health problems such as STIs, substance abuse, and mental illness; a devout Muslim may interpret illness as the will of Allah, a test of faith; may rely on ritual cures or alternative therapies before seeking help from health care provider; after death, the family may

want to prepare the body by washing and wrapping the body in unsewn white cloth; postmortem examinations are discouraged unless required by law. May avoid pork and alcohol if Muslim. Islamic patients observe month long fast of Ramadan (begins approximately mid-October); people suffering from chronic illnesses, pregnant women, breast-feeding, or menstruating don't fast. Females avoid eye contact with males; use same-sex family members as interpreters.

- **Asian Americans**—May value ability to endure pain and grief with silent stoicism; typically family oriented; extended family should be involved in care of dying patient; believes in “hot-cold” yin/yang often involved; sodium intake is generally high because of salted and dried foods; may believe prolonged eye contact is rude and an invasion of privacy; may not without necessarily understanding; may prefer to maintain a comfortable physical distance between the patient and the health care provider.
- **Latino Americans**—May view illness as a sign of weakness, punishment for evil doing; may consult with a curandero or voodoo priest; family members are typically involved in all aspects of decision making such as terminal illness; may see no reason to submit to mammograms or vaccinations.

- **Native Americans**—May turn to a medicine man to determine the true cause of an illness; may value the ability to endure pain or grief with silent stoicism; diet may be deficient in vitamin D and calcium because many suffer from lactose intolerance or don't drink milk; obesity and diabetes are major health concerns; may divert eyes to the floor when they are praying or paying attention.
- **Western Culture**—May value technology almost exclusively in the struggle to conquer diseases; health is understood to be the absence, minimization, or control of disease process; eating utensils usually consists of knife, fork, and spoon; three daily meals is typical.

20. Common Diets

- **Acute Renal Disease**—protein-restricted, high-calorie, fluid-controlled, sodium and potassium controlled.
- **Addison's disease**—increased sodium, low potassium diet.
- **ADHD and Bipolar**—high-calorie and provide finger foods.
- **Burns**—high protein, high caloric, increase in Vitamin C.
- **Cancer**—high-calorie, high-protein.
- **Celiac Disease**—gluten-free diet (no BROW: barley, rye, oat, and wheat).
- **Chronic Renal Disease**—protein-restricted, low-sodium, fluid-restricted, potassium-restricted, phosphorus-restricted.

- **Cirrhosis (stable)**—normal protein
- **Cirrhosis with hepatic insufficiency**—restrict protein, fluids, and sodium.
- **Constipation**—high-fiber, increased fluids
- **COPD**—soft, high-calorie, low-carbohydrate, high-fat, small frequent feedings
- **Cystic Fibrosis**—increase in fluids.
- **Diarrhea**—liquid, low-fiber, regular, fluid and electrolyte replacement
- **Gallbladder diseases**—low-fat, calorie-restricted, regular
- **Gastritis**—low-fiber, bland diet
- **Hepatitis**—regular, high-calorie, high-protein
- **Hyperlipidemias**—fat-controlled, calorie-restricted
- **Hypertension, heart failure, CAD**—low-sodium, calorie-restricted, fat-controlled
- **Kidney Stones**—increased fluid intake, calcium-controlled, low-oxalate
- **Nephrotic Syndrome**—sodium-restricted, high-calorie, high-protein, potassium-restricted.
- **Obesity, overweight**—calorie-restricted, high-fiver
- **Pancreatitis**—low-fat, regular, small frequent feedings; tube feeding or total parenteral nutrition.
- **Peptic ulcer**—bland diet
- **Pernicious Anemia**—increase Vitamin B12 (Cobalamin), found in high amounts on shellfish, beef liver, and fish.

- **Sickle Cell Anemia**—increase fluids to maintain hydration since sickling increases when patients become dehydrated.
- **Stroke**—mechanical soft, regular, or tube-feeding.
- **Underweight**—high-calorie, high protein
- **Vomiting**—fluid and electrolyte replacement

21. Positioning Clients

- **Asthma**—orthopneic position where patient is sitting up and bent forward with arms supported on a table or chair arms.
- **Post Bronchoscopy**—flat on bed with head hyperextended.
- **Cerebral Aneurysm**—high Fowler's.
- **Hemorrhagic Stroke**: HOB elevated 30 degrees to reduce ICP and facilitate venous drainage.
- **Ischemic Stroke**: HOB flat.
- **Cardiac Catheterization**—keep site extended.
- **Epistaxis**—lean forward.
- **Above Knee Amputation**—elevate for first 24 hours on pillow, position on prone daily for hip extension.
- **Below Knee Amputation**—foot of bed elevated for first 24 hours, position prone daily for hip extension.
- **Tube feeding for patients with decreased LOC**—position patient on right side to promote emptying of the stomach with HOB elevated to prevent aspiration.
- **Air/Pulmonary embolism**—turn patient to left side and lower HOB.
- **Postural Drainage**—Lung segment to be drained should be in the uppermost position to allow gravity to work.
- **Post Lumbar puncture**—patient should lie flat in supine to prevent headache and leaking of CSF.
- **Continuous Bladder Irrigation (CBI)**—catheter should be taped to thigh so legs should be kept straight.
- **After myringotomy**—position on the side of affected ear after surgery (allows drainage of secretion).
- **Post cataract surgery**—patient will sleep on unaffected side with a night shield for 1-4 weeks.
- **Detached retina**—area of detachment should be in the dependent position.
- **Post thyroidectomy**—low or semi-Fowlers, support head, neck and shoulders.
- **Thoracentesis**—sitting on the side of the bed and leaning over the table (during procedure); affected side up (after procedure).
- **Spina Bifida**— position infant on prone so that sac does not rupture.
- **Buck's Traction**—elevate foot of bed for counter-traction.
- **Post Total Hip Replacement**—don't sleep on operated side, don't flex hip more than 45-60 degrees, don't elevate HOB more than 45 degrees. Maintain hip abduction by separating thighs with pillows.
- **Prolapsed cord**—knee-chest position or Trendelenburg.
- **Cleft-lip**—position on back or in infant seat to prevent trauma to the suture line. While feeding, hold in upright position.
- **Cleft-palate**—prone.
- **Hemorrhoidectomy**—assist to lateral position.
- **Hiatal Hernia**—upright position.
- **Preventing Dumping Syndrome**—eat in reclining position, lie down after meals for 20-30 minutes (also restrict fluids during meals, low fiber diet, and small frequent meals).
- **Enema Administration**—position patient in left-side lying (Sim's position) with knees flexed.
- **Post supratentorial surgery (incision behind hairline)**—elevate HOB 30-45 degrees.
- **Post infratentorial surgery (incision at nape of neck)**—position patient flat and lateral on either side.
- **Increased ICP**—high Fowler's.
- **Laminectomy**—back as straight as possible; log roll to move and sand bag on sides.
- **Spinal Cord Injury**—immobilize on spine board, with head in neutral position. Immobilize head with padded C-collar, maintain traction and alignment of head

manually. Log roll client and do not allow client to twist or bend.

- **Liver Biopsy**—right side lying with pillow or small towel under puncture site for at least 3 hours.
- **Paracentesis**—flat on bed or sitting.
- **Intestinal Tubes**—place patient on right side to facilitate passage into duodenum.
- **Nasogastric Tubes**—elevate HOB 30 degrees to prevent aspiration. Maintain elevation for continuous feeding or 1 hour after intermittent feedings.
- **Pelvic Exam**—lithotomy position.
- **Rectal Exam**—knee-chest position, Sim's, or dorsal recumbent.
- **During internal radiation**—patient should be on bed rest while implant is in place.
- **Autonomic Dysreflexia**—place client in sitting position (elevate HOB) first before any other implementation.
- **Shock**—bed rest with extremities elevated 20 degrees, knees straight, head slightly elevated (modified Trendelenburg).
- **Head Injury**—elevate HOB 30 degrees to decrease intracranial pressure.
- **Peritoneal Dialysis when outflow is inadequate**—turn patient side to side before checking for kinks in the tubing.
- **Myelogram**
 - Water-based dye—semi Fowler's for at least 8 hours.

- Oil-based dye—flat on bed for at least 6-8 hours to prevent leakage of CSF.
- Air dye—Trendelenburg.

22. Common Signs and Symptoms

- **Pulmonary Tuberculosis (PTB)**—low-grade afternoon fever.
- **Pneumonia**—rust-colored sputum.
- **Asthma**—wheezing on expiration.
- **Emphysema**—barrel chest.
- **Kawasaki Syndrome**—strawberry tongue.
- **Pernicious Anemia**—red beefy tongue.
- **Down syndrome**—protruding tongue.
- **Cholera**—rice-watery stool and washer woman's hands (wrinkled hands from dehydration).
- **Malaria**—stepladder like fever with chills.
- **Typhoid**—rose spots in the abdomen.
- **Dengue**—fever, rash, and headache. Positive Herman's sign.
- **Diphtheria**—pseudo membrane formation.
- **Measles**—Koplik's spots (clustered white lesions on buccal mucosa).
- **Systemic Lupus Erythematosus**—butterfly rash.
- **Leprosy**—leonine facies (thickened folded facial skin).
- **Bulimia**—chipmunk facies (parotid gland swelling).
- **Appendicitis**—rebound tenderness at McBurney's point. Rovsing's sign (palpation of LLQ elicits pain in RLQ).

Psoas sign (pain from flexing the thigh to the hip).

- **Meningitis**—Kernig's sign (stiffness of hamstrings causing inability to straighten the leg when the hip is flexed to 90 degrees), Brudzinski's sign (forced flexion of the neck elicits a reflex flexion of the hips).
- **Tetany**—hypocalcemia, [+] Trousseau's sign; Chvostek sign.
- **Tetanus**—Risus sardonicus or rictus grin.
- **Pancreatitis**—Cullen's sign (ecchymosis of the umbilicus), Grey Turner's sign (bruising of the flank).
- **Pyloric Stenosis**—olive like mass.
- **Patent Ductus Arteriosus**—washing machine-like murmur.
- **Addison's disease**—bronzelike skin pigmentation.
- **Cushing's syndrome**—moon face appearance and buffalo hump.
- **Grave's Disease (Hyperthyroidism)**—Exophthalmos (bulging of the eye out of the orbit).
- **Intussusception**—Sausage-shaped mass.
- **Multiple Sclerosis**—Charcot's Triad: nystagmus, intention tremor, and dysarthria.
- **Myasthenia Gravis**—descending muscle weakness, ptosis (drooping of eyelids).
- **Guillain-Barre Syndrome**—ascending muscles weakness.

- **Deep vein thrombosis (DVT)**—Homan's Sign.
- **Angina**—crushing, stabbing pain relieved by NTG.
- **Myocardial Infarction (MI)**—crushing, stabbing pain radiating to left shoulder, neck, and arms. Unrelieved by NTG.
- **Parkinson's disease**—pill-rolling tremors.
- **Cytomegalovirus (CMV) infection**—Owl's eye appearance of cells (huge nucleus in cells).
- **Glaucoma**—tunnel vision.
- **Retinal Detachment**—flashes of light, shadow with curtain across vision.
- **Basilar Skull Fracture**—Raccoon eyes (periorbital ecchymosis) and Battle's sign (mastoid ecchymosis).
- **Buerger's Disease**—intermittent claudication (pain at buttocks or legs from poor circulation resulting in impaired walking).
- **Diabetic Ketoacidosis**—acetone breathe.
- **Pregnancy Induced Hypertension (PIH)**—proteinuria, hypertension, edema.
- **Diabetes Mellitus**—polydipsia, polyphagia, polyuria.
- **Gastroesophageal Reflux Disease (GERD)**—heart burn.
- **Hirschsprung's Disease (Toxic Megacolon)**—ribbon-like stool.
- Sexual Transmitted Infections:
 - **Herpes Simplex Type II**—painful vesicles on genitalia
 - **Genital Warts**—warts 1-2 mm in diameter.

- **Syphilis**—painless chancres
- **Chancroid**—painful chancres.
- **Gonorrhea**—green, creamy discharges and painful urination.
- **Chlamydia**—milky discharge and painful urination.
- **Candidiasis**—white cheesy odorless vaginal discharges.
- **Trichomoniasis**—yellow, itchy, frothy, and foul-smelling vaginal discharges.

23. Miscellaneous Tips

- Delegate sterile skills (e.g., dressing change) to the RN or LPN.
- Where non-skilled care is required, delegate the stable client to the nursing assistant.
- Assign the most critical client to the RN.
- Clients who are being discharged should have final assessments done by the RN.
- The **Licensed Practical Nurse (LPN)** can monitor clients with IV therapy, insert urinary catheters, feeding tubes, and apply restraints.
- Assessment, teaching, medication administration, evaluation, unstable patients cannot be delegated to an unlicensed assistive personnel.
- **Weight** is the best indicator of dehydration.
- When patient is in distress, **administration of medication is rarely the best choice.**
- Always check for allergies before administering antibiotics.

- **Neutropenic patients** should not receive vaccines, fresh fruits, or flowers.
- **Nitroglycerine patch** is administered up to three times with intervals of five minutes.
- **Morphine** is contraindicated in pancreatitis because it causes spasms of the Sphincter of Oddi. **Demerol** should be given.
- Never give **potassium (K+)** in IV push.
- Infants born to an HIV-positive mother **should receive all immunizations of schedule.**
- **Gravida** is the number of pregnancies a woman has had, regardless of outcome.
- **Para** is the number of pregnancies that reached viability, regardless of whether the fetus was delivered alive or stillborn. A fetus is considered viable at 20 weeks' gestation.
- **Lochia rubra** is the vaginal discharge of almost pure blood that occurs during the first few days after childbirth.
- **Lochia serosa** is the serous vaginal discharge that occurs 4 to 7 days after childbirth.
- **Lochia alba** is the vaginal discharge of decreased blood and increased leukocytes that's the final stage of lochia. It occurs 7 to 10 days after childbirth.
- In the event of fire, the acronym most often used is **RACE**. (R) Remove the patient. (A) Activate the alarm. (C) Attempt to contain the fire by closing

DISEASE

PATHOPHYSIOLOGY

SIGNS & SYMPTOMS

RISK FACTORS

DIAGNOSTIC TESTS

MEDICAL TREATMENT

COMPLICATIONS

NURSING INTERVENTIONS

TIME	MON	TUES	WED	THURS	FRI	SAT	SUN
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